

ARK Platform ATLAS PromptWare Design Document

Produced by **ATLAS ULTRA SI** (Automated Translation & Layout System) operating in **Mode 4 — LIVING PDD**: reverse-translation of the deployed ARK Platform monorepo (38 client routes · 136 API endpoints · 49 Drizzle tables · 22 feature flags · 19 migrations) into a certified 4-Part PromptWare Design Document, current through the merged **Training-Providers funnel + INDIVIDUAL_EXPLORER freemium tier**, the **Primitive Card Verification** engine, the one-page **ARK REPORT**, the promoted Intelligence-Hub tab navigation, and the **F1000 (First 1000) soft-launch promo**.

"Every requirement document is a blueprint in disguise. ATLAS reveals it."

We don't build AI agents. We engineer the DNA that governs them — powered by your cognition, owned by you.

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Card Invocation & Execution Log

Invocation: *Run ATLAS ULTRA SI on [DEPLOYED ARK PLATFORM CODEBASE] → produce 4-Part LIVING PDD of current status.* **GRO DNA:** LIFE MODE · **FORGE Step:** Stage 7 (Deployed → Documented) · **Mode:** 4 — Living PDD (reverse-translation).

This PDD does **not** propose a future build. It reverse-translates the *already-deployed* ARK Platform monorepo into the ATLAS 4-Part Standard, so any engineer, executive, or AI agent can pick up the current system without reading 170+ source files. Every Atomic Prompt in Part 3 maps to code that exists today. This revision (**v11**) supersedes v10 by absorbing the new **growth-and-monetization funnel** (Explorer freemium tier + Suggested Training Providers), the **Primitive Card Verification** flywheel surface, the one-page **ARK REPORT**, and the **F1000 (First 1000) soft-launch promo**.

Honesty Gate (threat_model G3)

ATLAS templates request speculative financials (**Estimated Annual ROI** , **Estimated Monthly Cost Savings**). Per the platform's **Honesty Gate G3** — *no fabricated ROI, no fake calculators, no NFT/Polygon claims, no external ideafactory.io billing* — those fields are reported as **NOT CLAIMED**. Token figures below are structural estimates of the document/prompt artifacts only, not revenue promises. The platform's real money surfaces are the Stripe-stub billing flow, the SPHINX credit ledger, and the **sponsored-placement + affiliate-click** monetization on the Training-Providers funnel — all documented as implemented, not projected.

Step 1 · Input Classification

ATTRIBUTE	VALUE
Input Type	Production Codebase (TypeScript monorepo) — routes via SPARTAN Dual-Input Codebase Path
Repository Shape	~44 server .ts modules · 130+ client .ts/.tsx modules · 1 shared schema/contract layer
Problem Statement	A 10-phase (A→J + funnel/verification increments) career-intelligence platform has accreted faster than its documentation; current status is legible only by reading source.

Solution Vision	A single certified 4-Part PDD that is the canonical, audience-segmented map of the deployed system, refreshed on every merged increment.
Success KPI	A reader reconstructs architecture, scoring math, route surface, funnel monetization, and Stage-1 gating in <15 minutes from this document alone.

Step 2 · Atomic Decomposition

The deployed surface decomposes into **6 capability domains**, each expressed as Atomic Prompts in Part 3:

1. **Foundation** — Express bootstrap, Drizzle/PG, session auth, security middleware.
2. **Identity & Scoring Core** — ARK/JST/CCMI engine, single-writer recalc, LHCS, SSE.
3. **Flywheel Engines** — Resume Analyzer, CCGE Arena, SPHINX Marketplace, Primitive Card Verification.
4. **AI Layer** — Anthropic Claude (Haiku KCSE / Sonnet narrative + scenarios), usage ledger, revenue guardrail.
5. **Growth & Monetization Funnel** — INDIVIDUAL_EXPLORER freemium tier, Suggested Training Providers (JST-matched ranking + sponsored placement + affiliate click tracking), one-page ARK REPORT, and the **F1000 (First 1000) soft-launch promo** (scarcity-bound founding-member acquisition — see Special Chapter).
6. **Edge & Governance** — Billing, GDPR, cohorts, institution workforce / HR connectors, GUIN+, Book Companion, feature-flag gating.

Step 3 · SPC Taxonomy Assignment

SPC	JCSE	CAMELOT SEAT	ROLE IN THIS PDD	LIVE IN PLATFORM
ATLAS ULTRA SI	50/50	Seat 3 — Transformation Architect	Lead PDD architect (this document)	<code>/marketplace</code> (200 cr)
SPARTAN SI	49/50	Seat 1 — Architecture	Codebase-path classifier (CLASS A/B/C)	<code>/marketplace</code> (175 cr)
BUGMXT SI	46/50	Seat 13 — Code Integrity	PDD-fidelity / drift auditor of the worksheet	<code>/marketplace</code> (150 cr)
SPHINX ULTRA SI	48/50	Seat 6 — Marketplace	Marketplace + credit-ledger + funnel lineage	Engine: <code>server/sphinx.ts</code>
ZPOS Expert SPC	47/50	Seat 10 — Resource	Token optimization of prompt artifacts	Engine: <code>server/zpos.ts</code>

PART 1 · Single-Page Cheat Sheet

Audience: All stakeholders · **Scan time:** <60 seconds

FIELD	VALUE
Document	ARK Platform — Living PDD · JNGL-ARK-PDD-CUR-2026-011 · v11 · 02 Jun 2026 · Status: DEPLOYED
Author	ATLAS ULTRA SI (SPC-ATLAS-001) · Transformation Architect · ATANDA Studio
JCSE Score	49/50 — Ultra Premium (FORGE Platinum)
Atomic Solution Summary	ARK is a full-stack AI career-intelligence platform that scores a person's automation resilience (ARK 0-600 = JST + CCMI), then drives a flywheel — resume analysis, a Context-Craft card game (CCGE), primitive verification quests, and a prompt marketplace (SPHINX) — that lets the score <i>grow</i> through verifiable skill activity. The dashboard now funnels users into a gated Suggested Training Providers surface (certifications ranked against the JST upskilling path), unlocked by a new free Explorer tier and monetized via sponsored placement + affiliate clicks. Everything beyond the public investor demo is gated behind login/subscription + feature flags.
Problem Statement	Knowledge workers cannot quantify their exposure to AI automation, see a concrete path to reduce it, <i>or</i> find the training that closes the exact gaps the engine surfaces.
Solution Vision	A single scored career identity (ARK) + a closed-loop flywheel that converts skill activity into audited score growth + a monetizable funnel from "here is your gap" to "here is who closes it."
Top 5 Features	• ARK Identity + JST/CCMI scoring engine with live SSE updates. • Resume Analyzer → 12-vector transferability + vulnerability assessment. • CCGE Arena + Primitive Card Verification — skill activity that <i>earns</i> score. • Suggested Training Providers — JST-ranked certifications, sponsored + affiliate-tracked, unlocked by the Explorer tier. • SPHINX Marketplace + 22-flag stage-gating holding CLASS-C surfaces until triggers fire.
Growth Surface	F1000 (First 1000) soft-launch promo — public-QR, scarcity-bound founding-member campaign: 1,000 single-use numbered codes → free Explorer entry + price-capped upgrade (\$10 PRO / \$9 SCHOOL). See Special Chapter.
Top 3 Benefits	• One auditable score (ARK) replaces a folder of disconnected assessments. • Score <i>increases</i> are earned and capped, so the number stays trustworthy. • The funnel turns a free assessment into two revenue surfaces (sponsored placement + affiliate) without leaving the product.
Token Savings	Raw source-to-prose ≈ 78,000 tokens → ZPOS-optimized PDD ≈ 39,000 tokens · ~50% reduction · Semantic preservation 97%

(artifact only)	
Production Cards	ATLAS 50/50 (Seat 3) · SPARTAN 49/50 (Seat 1) · BUGMXT 46/50 (Seat 13) · SPHINX 48/50 (Seat 6)
Stage-1 Flags ON	<code>investorDemo</code> · <code>executiveReport</code> · <code>cardVerification</code> · <code>trainingProviders</code> · <code>f1000Promo</code> (5 of 22)
Alignment Statement	This PDD satisfies the request <i>"Draft a Comprehensive Living PDD that captures the present ARK codebase using ATLAS format."</i>

PART 2 · Executive Summary

Audience: C-Suite, product owners, sponsors · **Format:** prose + tables

1. Business Context (The WHY)

The market gap ARK addresses is *quantified AI-career resilience*. Workers and institutions have access to resume tools and skills assessments, but none produce a single, longitudinal, defensible score that (a) measures exposure to automation, (b) responds to deliberate skill activity, and (c) routes the person to the training that closes their specific gaps. ARK closes that gap with the **ARK score (0-600)** — a composite of **JST** (Jobs-Skills-Talent, 0-300) and **CCMI** (Context Craft Mastery Index, 0-300) — surrounds it with a **flywheel** so the score is something a user *grows*, and now terminates the flywheel in a **monetizable funnel**: the Intelligence Hub's "Your Path" tab leads to **Suggested Training Providers**, where certifications are ranked against the engine's own upskilling recommendations.

The deployed system remains at **Stage 1 — MVP**, but the public surface has **widened** from the original 7 CLASS-A features. Four flags are now ON (**investorDemo** , **executiveReport** , **cardVerification** , **trainingProviders**), reflecting a deliberate push to (1) prove verifiable score growth and (2) stand up the revenue funnel. The remaining capability (cohorts, enterprise/institution workforce intelligence, corporate marketplace, advanced SPHINX, public GUIN+, Book Companion, etc.) is fully built but **feature-flagged off** behind documented commercial triggers, so each gate lifts without a redeploy.

2. Requirements Summary

PRIORITY	REQUIREMENT
MUST	A single, atomically-written score (ARK) with a <i>single writer</i> and an enforced invariant ($ARK = JST + CCMI \leq 600$).
MUST	Server-derived identity on every mutation; no client-supplied actor IDs; bcrypt at the storage boundary.
MUST	Flywheel caps that are strict ceilings (CCGE +15 ARK/day, SPHINX +20 ARK/30d, Verification +40 ARK/day) so the score cannot be farmed.
MUST	Stage gating: CLASS-C surfaces return 404 (indistinguishable from unimplemented) until their flag flips; the funnel is gated behind login + plan access.

MUST	Funnel integrity: affiliate outbound URLs validated to http(s)-only; provider self-registration lands in a pending state for admin approval; sponsorship boosts ranking <i>order</i> only, never the honest match score.
SHOULD	Live score updates via SSE; AI narrative/scenario generation under a token <i>and</i> cost budget.
SHOULD	GDPR export + cascade delete; auditable billing, credit, and affiliate-click ledgers; one-page exportable ARK REPORT (PDF/PNG/JPEG).
NICE	Institutional cohorts, enterprise/institution workforce intelligence, public contributor profiles, reader onboarding (Book Companion).

3. Features & Benefits Matrix

FEATURE	BENEFIT	KPI	BUSINESS VALUE
ARK Identity + scoring engine	One trustworthy career score	ARK invariant holds under cap scaling	Defensible IP; single source of truth
Live SSE updates (<code>useArkStream</code>)	Score reacts to activity in real time	<code>ark.identity</code> event on every flywheel write	Engagement / stickiness
Resume Analyzer	Automation-risk + 12-vector mobility map	Vulnerability level 0-4 per upload	Top-of-funnel value, free tier
CCGE Arena	Skill activity that <i>earns</i> score	+15 ARK/day cap respected	Daily-active driver
Primitive Card Verification	Evidence-gated proof of mastery	per-card badge; +40 ARK/day cap	Trust + credential signal
Suggested Training Providers	JST-matched certifications	sponsored boost = order only; affiliate click logged	New revenue funnel
Explorer freemium tier	Free unlock of the funnel	<code>trainingProviderAccess</code> true on Explorer	Top-of-funnel acquisition
ARK REPORT (<code>/report</code>)	One-page exec brief, exportable	PDF + PNG + JPEG export	Shareable artifact / virality
SPHINX Marketplace	Monetizable prompt IP + credits	70/30 split on transactional purchase	Revenue + creator network effects
F1000 soft-launch promo	Scarcity-bound founding-member acquisition	1,000 single-use codes; idempotent claim	Top-of-funnel growth engine

Feature-flag gating	Ship lean, scale without redeploy	22 flags; 5 ON in Stage 1	De-risked phased GTM
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4. SPC Taxonomy (Lineage)

SPC NAME	JCSE	CAMELOT SEAT	DISC	CATEGORY	VERSION
ATLAS ULTRA SI	50/50	3 — Transformation Architect	C+D	Innovation	1.0
SPARTAN SI	49/50	1 — Architecture	D+C	Innovation (Compression)	1.0.0
BUGMXT SI	46/50	13 — Code Integrity	C+D	Best Practice	1.0
SPHINX ULTRA SI	48/50	6 — Marketplace	I+D	Innovation	1.0
ZPOS Expert SPC	47/50	10 — Resource	C+S	Innovation	1.0

5. JCSE Score Breakdown (this PDD)

DIMENSION	MAX	SCORE	NOTE
Clarity	5	5	Audience-segmented, zero-ambiguity sections
Completeness	5	5	All 6 capability domains + full route/table counts
Coherence	5	5	FORGE 7-step → 4-Part flow preserved
Atomicity	5	5	Part 3 prompts are single-operation, I/O-typed
Token Efficiency	5	5	ZPOS ~50% reduction vs raw source-to-prose
Semantic Integrity	5	5	97% preservation; facts traced to code
Stakeholder Fit	5	5	Parts 1-4 target distinct audiences
Framework Alignment	5	5	FORGE + Context Craft + Camelot + VIBE
Executable Quality	5	4	Living PDD documents existing code; minor inferred metrics
IP Protection	5	5	Honesty Gate G3 enforced; lineage documented
TOTAL	50	49/50	Ultra Premium — FORGE Platinum

6. Risk & Mitigation Summary (Wasp Cards)

#	RISK	SEVERITY	MITIGATION (IN-CODE TODAY)
W1	Score farming via flywheel	High	Strict caps in <code>arkRecalc.ts</code> (CCGE +15/day, SPHINX +20/30d, Verification +40/day); overshoot hard-clamped off CCMI; manual/backfill can never award positive ARK.
W2	Privilege escalation (plan/cert/role)	High	<code>requireSelf</code> ; profile update strips <code>role</code> ; cert is 403/flywheel-only; admin gated by <code>ADMIN_USER_ID</code> (fail-closed).
W3	Unbounded AI cost	Med-High	Two-gate budget (token cap and cost cap); usage ledger; Haiku for cheap KCSE.
W4	Flagged surface leakage	Medium	<code>requireFeature</code> returns 404; client only registers a <code><Route></code> when flag is on (memory: re-check inside always-on parents — e.g. <code>/marketplace/:id</code>).
W5	PII / resume disclosure	Medium	Public DTOs only; SPC bodies locked until purchase; bcrypt hashes; helmet CSP + no-referrer.
W6	Funnel plan-gate bypass	Medium	Every self-serve provider/mutation route re-checks <code>hasTrainingAccess()</code> (FREE → 403), not just the consumer reads.
W7	Affiliate open-redirect / script URL	Medium	<code>/api/training/click</code> validates the outbound target to http(s) only ; <code>javascript:</code> and other schemes resolve to <code>url: null</code> .
W8	Sponsored ranking distorts honesty	Low-Med	Sponsorship boosts ordering only ; the displayed match score and reasons come from the unbiased JST token-set match in <code>trainingMatch.ts</code> .

7. PDD Metadata

FIELD	VALUE
Version	v11 (<code>JNGL-ARK-PDD-CUR-2026-011</code>)
Classification	Internal — Architecture of Record
Standard Alignment	PMBOK 7 · SAFe 6.0 · ISO/JNGL-4830:2026 · Context Craft 7-Pillar
Review Cadence	Refresh on each merged increment / Stage gate change
Supersedes	<code>ARK_PDD_Current_ATLAS</code> v10 (30 May 2026, through Task #23)

8. Bibliography & Data Sources

[replit.md](#) (architecture of record) · [threat_model.md](#) (G3 Honesty Gate, trust boundaries) · [shared/schema.ts](#) (49 tables + [SCORE_GLOSSARY](#) + [F1000_PROMO](#)) · [shared/featureFlags.ts](#) (22 flags) · [server/scoringEngine.ts](#) · [server/arkRecalc.ts](#) · [server/trainingMatch.ts](#) (funnel ranking) · [server/cardVerification.ts](#) · [server/billing.ts](#) (F1000 price-cap) · [server/routes.ts](#) (136 [/api/](#)* endpoints) · [migrations/0000-0018](#) · [exports/ARK_PDD_Current_MVP_Spartan.md](#) (Stage-1 scope) · the integrated SPC source files (ATLAS / BUGMXT / SPARTAN / SPHINX / ZPOS).

PART 3 · Comprehensive Worksheet (Atomic Prompts)

Audience: Technical team, AI agents, QA · **Format:** phased, color-coded, I/O-typed

Each row is an Atomic Prompt describing a capability **that exists in the deployed code**, with its real owning module. **Token Count** is an artifact estimate of the prompt text.

6-Phase Deployment Map

PHASE	NAME	FOCUS	BAND
1	Foundation	Express, Drizzle/PG, session auth, security	□ P0
2	Identity & Scoring Core	ARK/JST/CCMI, single-writer recalc, LHCS, SSE	□ P0 / □ P1
3	Flywheel Engines	Resume Analyzer, CCGE, SPHINX, Verification	□ P1
4	AI Layer	Claude KCSE/narrative/scenarios, guardrail	□ P1 / □ P2
5	Growth & Monetization Funnel	Explorer tier, Training Providers, ARK REPORT	□ P1 / □ P2
6	Edge & Governance	Billing, GDPR, cohorts, workforce, GUIN+, flags	□ P2 / □ P3

Phase 1 — Foundation

ID	PRI	DESCRIPTION (SINGLE OP)	INPUT	OUTPUT	VALIDATION
ARK-001	□ P0	Bootstrap Express on :5000 with helmet, rate limits, body limits	env	running server	/api/featu 200
ARK-002	□ P0	Provide single DB access surface via Drizzle	DATABASE_URL	DatabaseStorage (IStorage)	query round-
ARK-003	□ P0	Build PG-backed session middleware (cookie ark.sid)	SESSION_SECRET	session middleware	refuses start secret in pro

ARK-004	□ P0	Authenticate login; bcrypt verify + legacy rehash	<code>{email,password}</code>	session + user	wrong creds 401
ARK-005	□ P0	Authorize via <code>requireSelf(:param)</code> = session userId	route param	allow/deny	mismatch →

Phase 2 — Identity & Scoring Core

ID	PRI	DESCRIPTION	INPUT	OUTPUT	VALIDATION	DE
ARK-010	□ P0	Compute JST = $(J \cdot .30 + S \cdot .40 + T \cdot .30) \cdot 3$ (pure)	sub-scores	JST 0-300	unit-tested <code>test:scoring</code>	00
ARK-011	□ P0	Compute CCMI = weighted P1-P7 · 3 (pure)	pillar scores	CCMI 0-300	sum $\leq$ 300	00
ARK-012	□ P0	Derive ARK = JST + CCMI + ARK-ID hash	JST,CCMI	ARK 0-600 + id	ARK invariant	01
ARK-013	□ P0	Single-writer recalc: persist users + pillars + signals + history atomically	event	committed txn	invariant preserved under cap scaling	01
ARK-014	□ P1	Enforce flywheel caps as strict ceilings	delta, source	clamped delta	<code>delta $\leq$ intendedCap</code>	01
ARK-015	□ P1	Compute LHCS = $\text{round}(.35 \cdot \text{CPR} + .35 \cdot \text{MPS} + .30 \cdot \text{LCIS})$ + status band	signals	readiness + status	thresholds 70/40	01
ARK-016	□ P1	Emit <code>ark.identity</code> SSE on flywheel events	event	SSE frame	client receives snapshot+merge	01

Phase 3 — Flywheel Engines

ID	PRI	DESCRIPTION	INPUT	OUTPUT	VAL
ARK-020	□ P1	Parse resume (PDF/TXT $\leq 10\text{MB}$), keyword-score 6 categories	multipart file	assessment	emi <code>as</code>
ARK-021	□ P1	Detect automation risk → vulnerability level 0-4	parsed text	risk modifiers	14 i
ARK-022	□ P1	Assign archetype (Architect/Orchestrator/Conductor)	signals	readinessProfile	wei
ARK-023	□ P1	Deal CCGE session (5 cards)	<code>{userId,tier}</code>	session	star

ARK-024	□ P1	Finalize CCGE atomically → JCSE + capped ARK	session,hand	finalize txn	+15
ARK-025	□ P1	Generate evidence-gated verification quest from O*NET/SFIA/WEF map	{cardId}	quest spec	card ma
ARK-026	□ P1	Score + finalize verification → tier + per-card badge	prompts	tier B60/S70/G80/P90	+40 row
ARK-027	□ P1	HIVE precheck a listing; gate publish at HIVE ≥80 & CC_400+	draft	hive/kcse score	red:
ARK-028	□ P1	Execute purchase txn: lock credits, 70/30 split, body unlock	{listingId,buyer}	purchase	atoi boo

Phase 4 — AI Layer

ID	PRI	DESCRIPTION	INPUT	OUTPUT	VALIDATION	DEPS	OWN
ARK-030	□ P1	Initialize Anthropic client via Replit integration	env creds	client	/api/ai/status ok	001	serv
ARK-031	□ P1	Score CCGE hand with Haiku (KCSE rubric K-C-S-E)	hand	JCSE	bounded tokens/timeout	030	serv
ARK-032	□ P2	Generate resume narrative with Sonnet (Pro+)	assessment	narrative	gated claudeNarrative	030	serv
ARK-033	□ P2	Generate CCGE scenario with Sonnet (admin)	tier prompt	scenario	admin-gated	030	serv
ARK-034	□ P1	Enforce two-gate budget (token cap AND cost cap)	request	allow/deny	FREE/ENT caps absolute	030	reven serv

Phase 5 — Growth & Monetization Funnel

ID	PRI	DESCRIPTION	INPUT	OUTPUT	VALID
ARK-050	□ P1	Define INDIVIDUAL_EXPLORER free	plan registry	plan + limits	Explore funnel

		tier with <code>trainingProviderAccess</code>			
ARK-051	□ P1	Rank courses vs latest assessment (upskilling + pivots + transferability)	assessment	ranked courses + reasons	token-s (len≥2)
ARK-052	□ P1	Layer sponsored boost on ranking order only	ranked list	reordered list	match
ARK-053	□ P1	Serve directory + detail + <code>/suggested</code> (gated)	session	provider/course DTOs	<code>require</code> <code>require</code> <code>hasTr</code>
ARK-054	□ P1	Self-serve provider registration → pending state	<code>{provider}</code>	pending row	FREE → routes
ARK-055	□ P1	Record affiliate click → safe outbound URL	<code>{providerId,courseId}</code>	<code>{url}</code>	http(s)- <code>url:n</code>
ARK-056	□ P2	Admin approve/reject/sponsor provider	admin action	status change	admin-
ARK-057	□ P2	Render one-page ARK REPORT, export PDF/PNG/JPEG	identity	brief	flag <code>execu</code> Pro+

Phase 6 — Edge & Governance

ID	PRI	DESCRIPTION	INPUT	OUTPUT	VALIDATION	
ARK-060	□ P2	Stripe-stub checkout: create → complete → entitle plan	plan	billing event	audit row written	(
ARK-061	□ P2	GDPR export + cascade delete (<code>confirm:"DELETE"</code>)	session	export / purge	all owned rows removed	(
ARK-062	□ P2	Instructor cohorts: members, assignments, grades CSV	instructor	cohort data	<code>requireInstructor</code>	(
ARK-063	□ P3	Institution workforce + HR connector configs	admin	workforce data	flag <code>institutionWorkforce</code>	(
ARK-064	□ P3	Book Companion journey + <code>/b/:slug</code> QR resolver + ledger	reader	journey/badges	flag <code>bookCompanion</code>	(

ARK-065	□ P0	Gate every CLASS-C route: requireFeature → 404 if off	flag key	allow/404	env overlay flips flag	
ARK-066	□ P2	Seed canonical data incl. 4 providers/8 courses + live SPCs	dev POST	seeded rows	403 in prod	

Token Economics Dashboard (artifact)

TOKEN ECONOMICS DASHBOARD – PDD ARTIFACT (not platform revenue)

Raw Token Count (source-to-prose): ~78,000 tokens

ZPOS Methodology Applied: SYNTHESIS + PRISM

Optimized Token Count (this PDD): ~39,000 tokens

Reduction Achieved: ~50%

Semantic Preservation Score: 97%

Context Window Utilization: ~20% of 200K limit

Estimated Monthly Cost Savings: NOT CLAIMED (Honesty Gate G3)

Estimated Annual ROI: NOT CLAIMED (Honesty Gate G3)

PLATFORM AI BUDGET (real, in-code): two-gate (token cap AND cost cap)

- KCSE scoring → Claude Haiku (low cost, per-session)
- Narrative/Scenario → Claude Sonnet (Pro+ / admin gated)
- Guardrail: FREE & ENTERPRISE cost caps are absolute, not 10%-derived

PLATFORM REVENUE SURFACES (real, in-code):

- Stripe-stub billing (plan entitlement)
- SPHINX credit ledger (70/30 transactional split)
- Training funnel: sponsored placement + affiliate click ledger

PART 4 · VIBE DJ + VIBE Orchestra Implementation Plan

Audience: PM, delivery team, tool admins · **Format:** PMBOK 7 + SAFe 6.0 aligned

4A · VIBE DJ Tool Selection

The deployed project profile (single full-stack TS monorepo, AI-integrated, solo/small-team velocity on Replit) yields this **as-built** tool playlist:

PRIORITY	TOOL	ROLE	% EFFORT	RATIONALE (AS-BUILT)
Primary	Replit Agent	Build, refactor, review, deploy	70%	Native environment; workflows, secrets, DB, deploy all in one.
Primary	Anthropic Claude	KCSE scoring + narrative + scenarios	15%	Wired via <code>javascript_anthropic_ai_integrations</code> (Haiku + Sonnet).
Secondary	Drizzle Kit	Schema + migrations (0000-0017)	8%	Type-safe schema is the contract layer.
Secondary	md-to-pdf + Puppeteer	PDD rendering (this document)	5%	<code>scripts/renderPdd.ts</code> + <code>scripts/pddStyles.ts</code> .
Tertiary	Vite	Dev server + client build	2%	Served by Express in one process.

4B · VIBE Orchestra Coordination

PHASE	PRIMARY	SECONDARY	SYNC	CONFLICT RESOLUTION
Foundation	Replit Agent	Drizzle Kit	per-merge	<code>shared/schema.ts</code> is source of truth

Scoring Core	Replit Agent + Claude	Drizzle	per-task	<code>arkRecalc.ts</code> is the single writer
Flywheel	Replit Agent	Claude (KCSE)	per-task	orchestrator event bus arbitrates
Funnel	Replit Agent	—	per-task	<code>trainingMatch.ts</code> honesty: boost = order only
AI Layer	Claude	usage ledger	continuous	guardrail two-gate budget
Production	Replit Deploy	Autoscale	continuous	<code>SESSION_SECRET</code> required to start

4C · Camelot Roundtable Team Sheet (as-mapped)

SEAT	KNIGHT TITLE	HUMAN ROLE	AI SPC PAIR	RESPONSIBILITY IN ARK
0	Platform Sovereign	Lead Architect	ATLAS + ADA	Architecture governance, invariants
1	Strategic Architect	Backend Lead	SPARTAN SI	Schema, recalc, caps, storage
3	Transformation Architect	Senior Eng	ATLAS ULTRA SI	This PDD; reverse-translation
6	Marketplace Steward	Product	SPHINX ULTRA SI	SPHINX listings + credit ledger + funnel
7	Innovation Oracle	AI Eng	Claude (Haiku/Sonnet)	KCSE, narrative, scenarios
8	Process Guardian	QA	BUGMXT SI / HIVE	Drift audit, scoring tests, gates
10	Resource Master	Controller	ZPOS Expert	Token + cost budget
12	Vigilance Keeper	Security	GRO / threat_model	G3 Honesty Gate, authz, PII, funnel URL safety

4D · Project Management Plan (forward maintenance)

Because the system is **deployed**, the sprint plan is a *Stage-gate lift* plan: each future increment flips one feature flag once its trigger fires — no rebuild required.

SPRINT	FOCUS	TRIGGER TO LIFT	FLAG(S)
Maintenance	Keep ARK invariant + tests green	continuous	—

□ Lifted	Verifiable score growth + revenue funnel + founding-member growth	done	<code>cardVerification</code> , <code>trainingProviders</code> , <code>executiveReport</code> , <code>f1000Promo</code>
Lift 1	Public contributor profiles	≥25 creators	<code>guinPublic</code> , <code>notifications</code>
Lift 2	Marketplace network effects	≥100 listings	<code>sphinxAdvanced</code> , <code>corporateMarketplace</code>
Lift 3	Institutional SKU	first SCHOOL_STUDENT licence	<code>cohorts</code> , <code>enterpriseDashboard</code> , <code>institutionWorkforce</code>
Lift 4	PRO paid features	PRO billing live	<code>claudNarrative</code> , <code>subscriptionCancel</code> , <code>assessmentEmail</code>
Lift 5	Reader onboarding	book launch	<code>bookCompanion</code> , <code>contextCraftPage</code>
Lift 6	Cost governance	first \$1k MRR or 80% cap crossing	<code>revenueGuardrail</code>

PMI Knowledge-Area alignment: Scope → feature-flag gating (404 discipline); Quality → JCSE tests + HIVE publish gate + BUGMXT drift audit; Cost → two-gate AI budget; Risk → Wasp register (Part 2 §6); Communication → audience-segmented Parts 1-4; Stakeholder → Part 1 (all) / Part 2 (exec) / Part 3 (tech) / Part 4 (PM).

SAFe PI mapping: PI Planning = this PDD refresh on each merged increment; each *Lift* above is a Program Increment gated by a commercial trigger, executed by flipping

`FEATURE_<KEY>=true` with no code change.

Special Chapter · F1000 Soft-Launch Promo (First 1000)

Audience: Founders, growth, investors · **Status:** ☐ LIVE (flag `f1000Promo` ON) · **Surface:** public landing QR → `/f1000`

*Why a dedicated chapter: the F1000 ("First 1000") promo is ARK's deliberate cold-start acquisition mechanism — a scarcity-bound, single-use invite campaign that turns the public launch QR into 1,000 founding members on a free **Explorer** tier, with a price-capped upgrade path. It is the one Stage-1 surface whose entire reason for existing is top-of-funnel growth, so it earns its own chapter rather than a single worksheet row.*

SC.1 · Concept

The first **1,000** signed-in users who activate the shared launch QR each receive a **strictly numbered** founding code — the "*Nth of 1000*". Redemption lands the member on **INDIVIDUAL_EXPLORER** (free, funnel-unlocked) and brands the account `f1000_member = true`, which **price-caps any later upgrade** at promo rates and **raises the AI cost/token allowance** so founders experience the full product, not a throttled demo.

This is a soft launch by design: bounded supply (1,000), no paid spend required to enter, every entrant captured as a real account, and a built-in upgrade incentive — with **no fabricated ROI or "savings" figure** (Honesty Gate G3). The promo widens Stage-1 from 4 ON flags to **5** and is the first acquisition surface that sits *in front of* the existing growth-and-monetization funnel (capability domain 5).

SC.2 · Mechanics (as-built)

ELEMENT	IMPLEMENTATION	INTEGRITY PROPERTY
Supply	Strict <code>seq 1...1000</code> (<code>f1000_invites.seq</code> UNIQUE)	Hard cap; the 1001st claim returns <code>409 sold_out</code>
Token	<code>F1000-<4-digit seq>-<random hex></code> — e.g. <code>F1000-0042-7F3A9C2E1B6D</code>	Single-use & unguessable — the public QR cannot be drained by scripting
Binding	<code>user_id</code> UNIQUE — one code per	Claim is idempotent : re-claiming returns

	account	the same code (200), never a second
Allocation trigger	Authenticated activation of the QR	No code is minted for anonymous traffic
Member flag	<code>users.f1000_member</code> boolean	Drives promo pricing at checkout (<code>priceCentsForPlan(plan, f1000Member)</code>)

SC.3 · Founding-member pricing (price-cap, not discount math)

PLAN	STANDARD		AI COST ALLOWANCE	AI TOKEN CAP
INDIVIDUAL_EXPLORER	free	free (entry tier)	standard Explorer	standard
INDIVIDUAL_PRO	normal PRO price	\$10/mo	\$29/mo (2900¢)	5,000,000 tok/mo
SCHOOL_STUDENT	normal student price	\$9/mo	\$9/mo (900¢)	2,000,000 tok/mo

The promo *caps* the live plan price for F1000 members; it does **not** invent a discount or savings number (Honesty Gate G3). The raised allowances are real budget constants in `shared/schema.ts::F1000_PROMO`, scaled ~10× so the token gate never binds before the dollar gate.

SC.4 · Atomic Prompts (I/O-typed)

Each row maps to deployed code, consistent with Part 3's worksheet discipline.

ID	PRI	DESCRIPTION (SINGLE OP)	INPUT	OUTPUT	VALIDATION
ARK-058	□ P1	Allocate this account's single-use F1000 code (idempotent)	session	<code>{status, invite}</code>	<code>seq</code> 1...1000 unique; re-claim returns same code
ARK-059	□ P1	Serve live pool counter (claimed / remaining) — public	—	pool stats	<code>requireFeature("f1000Promo")</code>

ARK-060	□ P1	Report this account's F1000 standing	session	{member, invite, stats}	requireFeature + requireAuth
ARK-061	□ P1	Cap plan price + raise AI allowance for members at checkout	plan, f1000Member	promo price/budget	cap applied only when f1000Member true

SC.5 · Risk & Mitigation (Wasp Cards)

#	RISK	SEVERITY	MITIGATION (IN-CODE TODAY)
F1	QR drained by scripted / guessed claims	High	Server-minted complex tokens (seq + random hex); claim requires auth; f1000ClaimLimiter = 10/min
F2	One user farming many codes	High	user_id UNIQUE on f1000_invites ; claim is idempotent (same code returned)
F3	Over-allocation past 1,000	High	seq UNIQUE 1...1000; pool exhaustion → 409 sold_out , never a 1001st row
F4	Promo price abuse by non-members	Medium	Pricing cap keyed off users.f1000_member , set only by a real allocation
F5	Surface leaks before the launch window	Low	Whole surface gated by f1000Promo ; requireFeature → 404 when off; env overlay flips it

F1000 is a growth mechanism, not a revenue promise: bounded supply, honest price-caps, and an idempotent, abuse-resistant claim path — fully consistent with Honesty Gate G3.

Appendix · Living Delta (v10 → v11)

What this revision absorbed since v10 (30 May 2026, through Task #23):

AREA	CHANGE	SURFACE / FLAG
Tier	New free INDIVIDUAL_EXPLORER tier alongside existing plans; <code>trainingProviderAccess</code> limit added to every plan	<code>shared/schema.ts</code>
Funnel	Suggested Training Providers — JST-matched ranking (<code>server/trainingMatch.ts</code> , token-set match), self-serve registration (pending → admin approval), seeded 4 providers / 8 courses	<code>trainingProviders</code> ON · <code>/training</code> , <code>/training/p/:slug</code> , <code>/training/register</code>
Monetization	Sponsored/featured placement (order-only boost) + affiliate click tracking with http(s)-only outbound validation	<code>server/routes.ts</code> <code>/api/training/*</code>
Flywheel	Primitive Card Verification — evidence-gated quests (O*NET/SFIA/WEF), tiered badges, +40 ARK/day cap	<code>cardVerification</code> ON · <code>/api/verification/*</code>
Reporting	One-page ARK REPORT with PDF/PNG/JPEG export	<code>executiveReport</code> ON · <code>/report</code>
Growth	F1000 (First 1000) soft-launch promo — 1,000 single-use numbered codes via public QR → free Explorer entry + price-capped upgrade (\$10 PRO / \$9 SCHOOL, raised AI allowance); idempotent, abuse-resistant claim	<code>f1000Promo</code> ON · <code>/f1000</code> , <code>/api/f1000/*</code>
UX	Intelligence-Hub tab bar promoted to top, numbered 1/2/3, throbs on inactive tabs, hover tooltips, <code>prefers-reduced-motion</code> fallback	<code>client/src/pages/dashboard.tsx</code>
Schema	Migrations grew 0009→0018 (guest assessments, cumulative profile, institution workforce, HR connectors, report bio/shares, CCGE custom card, card verifications, training providers, F1000 promo)	<code>migrations/</code>

Metrics	33→ 38 client routes · 99→ 136 API endpoints · 37→ 49 tables · 18→ 22 flags · 9→ 19 migrations · 1→ 5 flags ON	platform-wide
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ATLAS compression complete. 4-Part LIVING PDD v11 delivered. JCSE 49/50 — FORGE Platinum. Honesty Gate G3 enforced: no fabricated ROI, no external billing, no NFT claims. Funnel monetization documented as implemented (sponsored placement + affiliate ledger), not projected.